

BEDROCK ESG — TECHNICAL BIBLE v2.4

(Trinity Protocol)

Classification: LLM / Developer Reference — Exhaustive Technical Specification
Backtrace: BT-C9C4C5 v2.4 | **Protocol:** Aethexer v1 | **Updated:** 2026-06-16
Domain: `bedrockesg.com` | **Cluster:** Solana Devnet | **Blockchain:** Hedera + Polygon (production attestation)

TABLE OF CONTENTS

- 1. [System Identity](#)
 - 2. [Architecture Overview](#)
 - 3. [Directory Structure](#)
 - 4. [Database Schema](#)
 - 5. [Authentication & RBAC](#)
 - 6. [Aethexer Protocol Layer](#)
 - 7. [Sentinel Engine](#)
 - 8. [Forensic Ingest Pipeline](#)
 - 9. [Telemetry Ingest Pipeline](#)
 - 10. [Discovery Pipeline](#)
 - 11. [Settlement Engine \(Universal Law\)](#)
 - 12. [Blockchain Layer](#)
 - 13. [Spatial Heritage Parser](#)
 - 14. [Carbon Calculation Engines](#)
 - 15. [Emission Factor Database](#)
 - 16. [HUM Dynamic Listener](#)
 - 17. [Admin Control Panel](#)
 - 18. [API Route Index](#)
 - 19. [Page Route Index](#)
 - 20. [Library Module Index](#)
 - 21. [Environment Variables](#)
 - 22. [Deployment](#)
-

1. SYSTEM IDENTITY

Key	Value
Platform Name	Bedrock ESG (formerly Scrapnet 2.0)
Protocol Version	Aethexer v1 — Trinity Protocol
Backtrace	BT-C9C4C5
Service Identity	9tsq8qB4P9uPRSHr-jXLaDQJd93nuo4wEEtkR1tAJRAQ3 (Aethexer Notary Node)
Founder Alias	Kato / Zer0 / Aethexer Notary Node
Admin Email	idecade8@gmail.com (ADMIN_MASTER_EMAIL)
Production URL	https://bedrockesg.com
Design Theme	Deep Space Dark Mode (#0a0a0a bg, #1a1a1a cards, electric blue accent)

2. ARCHITECTURE OVERVIEW

Codebase Statistics

Metric	Count
Prisma Models	45
Prisma Enums	30
API Routes (route.ts)	76
Pages (page.tsx)	31
Components (.tsx)	99
Library Modules (lib/*.ts)	31
Schema Lines	2,041
universal-law.ts Lines	2,011
forensic-ingest/route.ts Lines	588
sentinel-engine.ts Lines	303

Core Layer Stack

PRESENTATION – 31 Pages + 99 Components (Dark Theme)
API LAYER – 76 route.ts handlers (v1 + legacy)
PROTOCOL – Aethexer Auth + Sentinel Engine + RBAC
SETTLEMENT – Universal Law (70/20/10 Zero Greed)
BLOCKCHAIN – Solana Devnet + Hedera HCS + Polygon
PERSISTENCE – PostgreSQL (45 models) + S3 Cloud Storage

3. DIRECTORY STRUCTURE

```

nextjs_space/
├── app/
│   ├── api/
│   │   ├── v1/
│   │   │   ├── forensic-ingest/
│   │   │   │   └── Aethexer Protocol v1 endpoints
│   │   │   ├── telemetry-ingest/
│   │   │   │   └── Unified forensic ingest (588 lines)
│   │   │   ├── telemetry-reevaluate/
│   │   │   │   └── IoT sensor ingest
│   │   │   ├── discovery/
│   │   │   │   └── Batch re-evaluation
│   │   │   └── admin/
│   │   │       ├── Macro-asset discovery CRUD
│   │   │       ├── purge-test-artifacts/
│   │   │       │   └── Destructive cleanup gate
│   │   │       ├── submissions/[id]/approve/
│   │   │       │   └── Founder approval gate
│   │   │       ├── auth/
│   │   │       │   └── Login, signup, Google SSO, JWT
│   │   │       ├── biochar/
│   │   │       │   └── CDR calculator + batches
│   │   │       ├── submissions/
│   │   │       │   └── Full submission lifecycle
│   │   │       ├── vault/
│   │   │       │   └── Verification repository
│   │   │       └── ...
│   │   │           └── 76 total route files
│   │   └── admin/
│   │       ├── Admin control panel
│   │       ├── discovery/
│   │       │   └── Macro-asset discovery UI
│   │       ├── submission-portal/
│   │       │   └── Corporate gatehouse
│   │       ├── onboarding/
│   │       │   └── 3-step sovereign onboarding
│   │       ├── vault/
│   │       │   └── Verification repository UI
│   │       ├── biochar/
│   │       │   └── CDR engine pages
│   │       ├── metal/
│   │       │   └── Metal recovery pages
│   │       ├── global-ledger/
│   │       │   └── Consolidated ledger view
│   │       ├── ledger/
│   │       │   └── Legacy ledger
│   │       ├── impact/
│   │       │   └── Impact dashboard
│   │       ├── legacy-healer/
│   │       │   └── Friction healing calculator
│   │       └── page.tsx
│   │           └── Main dashboard + cockpit ticker
│   └── components/
│       ├── cockpit-ticker.tsx
│       │   └── Real-time asset ticker (SSE)
│       ├── discovery-dashboard.tsx
│       │   └── Discovery feed UI
│       ├── sidebar.tsx
│       │   └── Navigation (incl. Admin section)
│       └── ...
│           └── 99 total component files
├── lib/
│   ├── aethexer-auth.ts
│   │   └── Trinity 3-layer auth guard
│   ├── aethexer-protocol.ts
│   │   └── Protocol envelope wrapper
│   ├── sentinel-engine.ts
│   │   └── 95/75/75 scoring engine
│   ├── universal-law.ts
│   │   └── Zero Greed settlement (2,011 lines)
│   ├── rbac.ts
│   │   └── 4-tier role hierarchy + permissions
│   ├── rbac-guard.ts
│   │   └── JWT extraction + route protection
│   ├── audit-logger.ts
│   │   └── Fire-and-forget audit pipeline
│   ├── validator.ts
│   │   └── Zod schemas for validation
│   ├── logistics-auditor.ts
│   │   └── Haversine distance + CO2 avoided
│   ├── emission-factors.ts
│   │   └── ICE v4.1 + Ecoinvent (40 factors)
│   ├── blockchain.ts
│   │   └── Hedera + Polygon attestation
│   ├── solana-notary.ts
│   │   └── Devnet memo anchoring
│   ├── solana-integration.ts
│   │   └── Wallet adapter integration
│   ├── ply-parser.ts
│   │   └── PLY/Splat spatial parser
│   ├── carbon-calculator.ts
│   │   └── Eurobitume LCA A1-A3
│   ├── metal-calculator.ts
│   │   └── Metal recovery calculations
│   ├── green-alpha-engine.ts
│   │   └── Green scoring engine
│   ├── consent.ts
│   │   └── SHA-256 consent hashes
│   ├── s3.ts
│   │   └── AWS S3 presigned URLs
│   ├── db.ts
│   │   └── Prisma client singleton
│   └── services/
│       ├── hum-listener.ts
│       │   └── Dynamic HUM baseline (12.6 MW)
│       └── prisma/
│           ├── schema.prisma
│           │   └── 45 models, 30 enums (2,041 lines)
│           ├── contexts/
│           │   └── React contexts (sidebar, auth)
│           ├── hooks/
│           │   └── Custom hooks
│           └── docs/
│               └── Project Bible documents

```



4. DATABASE SCHEMA

Model Registry (45 Models)

Model	Table	Purpose
User	users	Core identity — email, password, role, wallet
InvitationCode	invitation_codes	Agent onboarding codes
FarmerProfile	farmer_profiles	Biochar farmer extended data
RealtorProfile	realtor_profiles	179D energy track data
BusinessProfile	business_profiles	Corporate entity data
SovereignProfile	sovereign_profiles	Sovereign individual data
UserCarbonEntry	user_carbon_entries	Per-user carbon footprint
Location	locations	Unified location model (replaced Yards)
Project	projects	RAP asphalt projects
Farmer	farmers	Biochar feedstock suppliers
BiocharBatch	biochar_batches	CDR production batches
RecyclingLog	recycling_logs	Material recycling records
FarmerCredit	farmer_credits	Green credit lifecycle
UserConsent	user_consents	Carbon rights + AI consent
ConsentLog	consent_logs	Consent audit trail
MetalRecoveryLog	metal_recovery_logs	Precious metal recovery
VaultNode	vault_nodes	Verification nodes
VaultNodeOperator	vault_node_operators	Node operators
VaultVerification	vault_verifications	Logistics audits
ComplianceFramework	compliance_frameworks	Regulatory frameworks
InternationalTransaction	international_transactions	Cross-border compliance
TravelRuleExchange	travel_rule_exchanges	FATF Travel Rule
VatTaxRate	vat_tax_rates	Jurisdiction tax rates

Model	Table	Purpose
PortfolioOffset	portfolio_offsets	Carbon offset portfolio
RetiredCredit	retired_credits	Retired carbon credits
FinancedEmissionLog	financed_emission_logs	PCAF financed emissions
UniversalLedgerEntry	universal_ledger_entries	Consolidated ledger
DeadMassExtraction	dead_mass_extractions	Legacy friction extraction
ResonancePrint	resonance_prints	HUM resonance records
SocialYield	social_yields	Social impact yields
PublicResilienceGrant	public_resilience_grants	10% public resilience allocations
ImpactReport	impact_reports	Impact assessment reports
LegacyAudit	legacy_audits	Historical audit records
VibrationalEquity	vibrational_equity	HUM equity state
ParametricInsurance	parametric_insurance	Insurance products
PulseString	pulse_strings	HUM pulse data
Submission	submissions	Core forensic submission (multi-track)
SubmissionDocument	submission_documents	File attachments per submission
Settlement	settlements	70/20/10 settlement records
UserProfile	user_profiles	Extended user profiles
SolanaAuditProof	solana_audit_proofs	On-chain anchor records
IngestNonce	ingest_nonces	Replay protection (24h TTL)
DiscoveryAsset	discovery_assets	Macro-asset discovery queue
AuditLog	audit_logs	System event audit trail (SHA-256)
TelemetryReading	telemetry_readings	IoT sensor telemetry

Key Enums

Enum	Values
UserRole	FOUNDER, ADMIN, SOVEREIGN_AGENT, CLIENT_OWNER, AGENT, FARMER, REALTOR, BUSINESS_OWNER, SOVEREIGN_INDIVIDUAL
SubmissionTrack	TRACK_A_MATERIALS, TRACK_B_ENERGY_179D, TRACK_C_BIOCHAR_CDR
SubmissionStatus	DRAFT, PENDING_FORENSIC_VERIFICATION, FORENSIC_VERIFIED, AUTO_APPROVED, PENDING_SOVEREIGN_REVIEW, PENDING_AUTHORIZATION, SOVEREIGN_HOLD, ESCROW_REVIEW, PENDING_PLATFORM_REVIEW, REJECTED, SETTLEMENT_COMPLETE
SettlementAssetClass	SOVEREIGN_SKIN, PROPERTY_INDUSTRIAL, PROPERTY_RESIDENTIAL, PHYSICAL_COMMODITIES, PRECIOUS_METALS_CUSTODIAL
DiscoveryTier	PENDING_SOVEREIGN_REVIEW, AUTO_APPROVED
DiscoveryCategory	EXCHANGE, SKYSCRAPER, CASINO
AssayStatus	UNASSAYED, PENDING, CERTIFIED

5. AUTHENTICATION & RBAC

Authentication Flow

Custom JWT system — NOT NextAuth. Cookie-based `auth-token` `httpOnly`.

Login Flow:

POST /api/auth/login → validates credentials → issues JWT → sets auth-token cookie

Google SSO Flow:

GET /api/auth/google → CSRF state token → redirect to Google OAuth

GET /api/auth/callback/google → exchange code → find/create user → issue JWT

Session Check:

GET /api/auth/me → reads auth-token cookie → returns { user: { id, email, name, role } }

Logout:

POST /api/auth/logout → clears auth-token cookie

Google SSO Notes:

- OAuth Client: “Bedrock ESG Production Gate” (933406767966-cfju...)
- Scoped to bedrockesg.com
- ADMIN_MASTER_EMAIL match → auto-elevated to ADMIN role
- New SSO users default to SOVEREIGN_INDIVIDUAL role

RBAC Hierarchy (lib/rbac.ts)

```
FOUNDER (100)    → system:full, founder:settings, audit_logs:export, roles:manage +
ALL below
ADMIN (75)       → platform:manage, users:manage, discovery:manage, notary:anchor + A
LL below
SOVEREIGN_AGENT → submissions:approve, submissions:verify, discovery:view, tele-
metry:ingest + ALL below
CLIENT_OWNER     → submissions:create, submissions:view_own, telemetry:view
```

Legacy role mapping: AGENT → SOVEREIGN_AGENT; FARMER/REALTOR/BUSINESS_OWNER/
SOVEREIGN_INDIVIDUAL → CLIENT_OWNER

JWT Guard (lib/rbac-guard.ts)

```
extractUser(req: NextRequest): { id, email, name, role } | null
withRbac(handler, requiredPermission): NextResponse
```

Dual Auth Pattern (v1 endpoints)

All /api/v1/* endpoints accept EITHER:

1. Aethexer protocol key (x-aethexer-key header) — for EXITZ field app
2. Browser session JWT (auth-token cookie) — for admin panel / web UI

6. AETHEXER PROTOCOL LAYER

Auth Guard — Trinity 3-Layer (lib/aethexer-auth.ts , 146 lines)

Layer	Mechanism	Threshold
1. Dual-Key	x-aethexer-key validated via crypto.timingSafeEqual() against AETHEX-ER_PROTOCOL_KEY OR AETHEX-ER_FIELD_KEY	Exact match
2. Timestamp Window	x-aethexer-ts header must be ±300 seconds of server time	300s
3. Nonce Dedup	x-aethexer-nonce stored in IngestNonce model; replays rejected	24h TTL

CRITICAL: validateAethexerKey() is **async** — all callers MUST await .

Protocol Envelope (lib/aethexer-protocol.ts , 74 lines)

Every /api/v1/* response wraps in:

```
{
  "protocol": "aethexer-v1",
  "backtrace": "BT-xxxxxxx",
  "timestamp": "2026-06-16T...",
  "sentinel": "NODE-01",
  "data": { ... },
  "status": 200
}
```

Functions: protocolSuccess(data, backtrace, status) , protocolError(code, message, status, backtrace) , generateBacktrace()

7. SENTINEL ENGINE (lib/sentinel-engine.ts , 303 lines)

Three-Pillar Scoring

Pillar	Threshold	Weight	Input Type
Document Integrity	≥ 0.95 (95%)	40%	DocumentIntegrityInput[] — per-file SHA-256 hash, magic bytes, header validation
LLM Confidence	≥ 0.75 (75%)	30%	LLMConfidenceInput — parser confidence, fields extracted vs expected, anomaly flag
Spatial Correlation	≥ 0.75 (75%)	30%	SpatialCorrelationInput — geolocation, timestamp consistency, metadata density, scan/photo presence

Verdict Matrix

Verdict	Condition
AUTO_APPROVED	All 3 pillars $\geq$ threshold
PENDING_SOVEREIGN_REVIEW	1 pillar below threshold
SOVEREIGN_HOLD	2 pillars below threshold
REJECTED	All 3 below threshold or composite < 0.50
PENDING_AUTHORIZATION	Edge case routing
ESCROW_REVIEW	Financial review required
PENDING_PLATFORM_REVIEW	Platform team review

Magic Byte Validation

Format	Signature
PDF	%PDF (0x25504446)
JPEG	FF D8 FF
PNG	89 50 4E 47 (.PNG)
XLSX/DOCX	50 4B (PK ZIP)
PLY	p ly \n (ASCII header)
Splat	≥ 32 bytes (binary blob)

Score Hash

SHA-256 of `${docScore}|${llmScore}|${spatialScore}|${verdict}|${timestamp}` — tamper-detection for audit trail.

8. FORENSIC INGEST PIPELINE (`/api/v1/forensic-ingest` , 588 lines)

Accepted Payload (multipart/form-data)

Field	Type	Required	Notes
<code>track</code>	string	✓	TRACK_A_MATERIALS / TRACK_B_ENERGY_179D / TRACK_C_BIOCHAR_CD
<code>walletAddress</code>	string	✓	Solana wallet address
<code>documents[]</code>	File[]	✓	PDF, CSV, XLSX, DOCX (max 100MB each)
<code>spatialScan</code>	File	✗	PLY or Splat file
<code>groundPhotos[]</code>	File[]	✗	JPG, PNG, WebP
<code>companyName</code>	string	✗	Corporate identity
<code>assetMetadata</code>	JSON string	✗	Must contain <code>latitude</code> / <code>longitude</code> for Haversine check
<code>operatorId</code>	string	✗	SPEXTER operator identifier
<code>facilityId</code>	string	✗	Facility zone ID (e.g. DC-SOVEREIGN-F1) — triggers Haversine boundary check
<code>clientTimestamp</code>	string/int	✗	ISO-8601 or Unix epoch ms — validated against server time (15min drift)
<code>sensorSnapshots</code>	JSON array	✗	SPEXTER sensor readings [<code>{sensorId, type, value, unit, capturedAt}</code>]

7-Step Pipeline

```

Step 1: Auth Gate (Trinity 3-Layer)
Step 2: Field Extraction + SPEXTER Extension Parsing
  [ ] operatorId, facilityId, clientTimestamp, sensorSnapshots
  [ ] Temporal spoofing detection (15min drift tolerance)
  [ ] Coordinates elevated from assetMetadata.latitude/longitude
Step 3: File Processing + Magic Byte Validation
  [ ] SHA-256 hash per file
  [ ] Magic byte check per buffer
  [ ] S3 presigned upload URL generation
  [ ] Hard reject on ANY magic byte mismatch (HTTP 422)
Step 3.5: HAVERSINE BOUNDARY ENFORCEMENT
  [ ] facilityId resolved via longest-prefix match against FACILITY_ZONE_REGISTRY
  [ ] Raw Haversine distance computed (R=6371km)
  [ ] Hard reject if distance > zone radius (HTTP 400)
  [ ] Error: "GPS ACCURACY DEGRADED – PLEASE RE-ENTER THE ACTIVE CAPTURE ZONE"
Step 4: Sentinel Engine Scoring
  [ ] DocumentIntegrityInput from file hashes
  [ ] LLMConfidenceInput enriched with sensor count + anomaly flag
  [ ] SpatialCorrelationInput enriched with timestamp consistency + sensor bonus
Step 5: Submission + Document Creation (Prisma)
  [ ] DRAFT status, backtrace ID, sentinel score in parserResultJson
  [ ] operatorId, facilityId, clientTimestamp, sensorSnapshots, haversineCheck persisted
Step 6: Protocol Response
  [ ] submissionId, files with upload URLs, sentinel verdict, spexterExtensions block

```

Facility Zone Registry

Prefix	Zone	Center	Radius
DC-SOVEREIGN	Sovereign Primary — Las Vegas NV	36.1699, -115.1398	50km
DC-PROPERTY	Property Industrial — Houston TX	29.7604, -95.3698	75km
DC-PALLETS	Logistics Hub — Ontario CA	34.0633, -117.6509	40km
DC-METALS	Metals Recovery — Pittsburgh PA	40.4406, -79.9959	60km

Haversine Implementation (lib/logistics-auditor.ts)

```

calculateDistance(origin: LogisticsCoordinates, destination: LogisticsCoordinates):
number
// R = 6371km, applies 1.2× route factor
// Forensic-ingest divides by 1.2 for raw boundary check

```

State Flow

DRAFT → PENDING_FORENSIC_VERIFICATION → FORENSIC_VERIFIED → [founderApproved] → SETTLEMENT_COMPLETE

9. TELEMETRY INGEST PIPELINE (/api/v1/telemetry-ingest)

Accepts IoT sensor data from DC nodes.

- **KNOWN_GEOGRAPHIC_ZONES** — Same 4 zones as Facility Zone Registry (DC-SOVEREIGN, DC-PROPERTY, DC-PALLETS, DC-METALS)
- **GPS derivation** from zone center when node prefix matches
- **Grid-intensity correlation** — validates reported intensity against zone range
- **Batch re-evaluation** — /api/v1/telemetry-reevaluate re-scores existing readings
- **Model:** TelemetryReading — stores nodeId, sensorType, value, coordinates, sentinelVerdict

10. DISCOVERY PIPELINE

Endpoints

Method	Endpoint	Action
GET	/api/v1/discovery	Fetch all discovery assets (filterable by verdict, category)
POST	/api/v1/discovery	Actions: approve , reject , rescan , settle

Model: DiscoveryAsset

Field	Type	Notes
symbol	String (unique)	NYSE-01, BURJ-DXB-001, MGM-LV-01, etc.
category	DiscoveryCategory	EXCHANGE, SKYSCRAPER, CASINO
estimatedValueUsd	Float	Estimated asset value
sovereignShare70	Float	70% of estimated value
verdict	DiscoveryTier	PENDING_SOVEREIGN_REVIEW / AUTO_APPROVED
settlementId	String?	DISC-* settlement ID when settled

Seeded Assets (12)

Exchanges (5): NYSE-01, NASDAQ-01, LSE-01, CME-01, EUREX-01

Skyscrapers (4): BURJ-DXB-001, TAI-101-TPE, OWT-NYC-001, SHA-TWR-001

Casinos (3): MGM-LV-01, VEN-MAC-01, MBS-SGP-01

11. SETTLEMENT ENGINE — UNIVERSAL LAW (lib/universal-law.ts , 2,011 lines)

Zero Greed Policy — 70/20/10 Split

```

UNIVERSAL_SPLIT = {
  FOUNDER_YIELD: 0.70,      // "Asset Sovereign" – Verified Asset Holder
  STEWARDSHIP: 0.20,       // "Platform Processor" – Bedrock Treasury
  PUBLIC_RESILIENCE: 0.10,  // "Public Resilience" – Community allocation
}
```

Runtime validation: FOUNDER_YIELD + STEWARDSHIP + PUBLIC_RESILIENCE must equal 1.0 (throws on violation).

Asset Class Variants (A-E)

Code	Variant	Description
A	SOVEREIGN_SKIN	Personal carbon / lifestyle
B	PROPERTY_INDUSTRIAL	Commercial real estate + industrial
C	PROPERTY_RESIDENTIAL	Residential real estate
D	PHYSICAL_COMMODITIES	Materials, recycling, logistics
E	PRECIOUS_METALS_CUSTODIAL	Gold, silver, platinum custody

Precious Metals VAL Formula

$$\text{VAL} = (\text{Spot Price} \times \text{Weight} \times \text{Purity\%}) - \text{Refinery Discount}$$

Fields on Submission: `purityPercentage`, `assayStatus`, `refineryDiscount`, `spotPriceAtIngest`, `verifiedAssetValue`

Settlement Model

Fields: `founderYieldUsd`, `stewardshipUsd`, `publicResilienceUsd`, `carbonTonnes`, `carbonValueUsd`, `assetClass`, `solanaSignature`, `solanaAnchoredAt`, `anchorPending`

12. BLOCKCHAIN LAYER

Solana Devnet Notary (`lib/solana-notary.ts`)

- Memo program: `MemoSq4gqABAXKb96qnH8TysNcWxMyWCqXgDLGmfChR`
- Payload: `{ bt, submissionId, action, timestamp }`
- Graceful fallback: no secret key → OFFLINE hash + PENDING_ANCHOR

EXITZ Token Ecosystem (Mainnet)

Token	Mint Address	Supply
EXITZ (base)	<code>44SrHT9Qwyz2jiiJkRF1zHKF2N-FTgTC8m6ubPTb5qJBJ</code>	2
EXITZ-C (Classic Vehicles)	<code>7ZPsJE3W5YXnnYgapzfsA5Jx-tK8VQwtTn5Jb23hgFzMK</code>	1
EXITZ-A (Fine Art)	<code>G11hBknKcBXUjc4kvAtAkw-joHz5yMr7FTqEEYC96EnJn</code>	1

Token-2022 program with on-chain metadata and forensic memo anchors.

Hedera + Polygon (`lib/blockchain.ts`)

- `executeDualBlockchainAttestation()` — Hedera HCS submission + Polygon ERC-1155 minting
 - Cross-chain metadata linking Hedera transaction ID to Polygon token URI
 - Called during batch verification in agent portal
-

13. SPATIAL HERITAGE PARSER (`lib/ply-parser.ts`)

- **PLY ingest:** ASCII + Binary Little-Endian
 - **Gaussian Splat detection:** SH coefficients, opacity, scale, rotation
 - **Volume → material mass → carbon estimate pipeline**
 - **Material densities:** asphalt 2.4 t/m³, aggregate 1.6 t/m³, concrete 2.3 t/m³
 - **API:** `/api/spatial` (GET status, POST parse)
-

14. CARBON CALCULATION ENGINES

Asphalt RAP Engine (`app/api/calculate/route.ts`)

- Local Eurobitume A1-A3 LCA formulas
- Virgin baseline: 60 kg CO₂/tonne
- RAP reduction: 0.4 kg/tonne per % RAP
- SSE streaming for real-time results

Biochar CDR Engine (`app/api/biochar/route.ts`)

- Pyrolysis temperature validation
- Stability class calculation
- CDR credit generation + Hedera Guardian compliance
- Agent commission system

Metal Recovery Engine (`lib/metal-calculator.ts`)

- Scrap metal carbon avoidance calculations
- Multiple metal types supported

Legacy Healer (`lib/legacy-healer.ts`)

- Friction extraction from dead-weight assets
 - Market rate multiplier calculations
 - Integrated with Universal Law 70/20/10 split
-

15. EMISSION FACTOR DATABASE (`lib/emission-factors.ts`)

- **ICE Database v4.1:** 25 factors
- **Ecoinvent v3.10.1:** 15 factors
- **Total:** 40 emission factors

- **Biogenic Variance Correction:** 60% active for US and BR markets
- **Grid mixes:** US (0.419), BR (0.082), EU (0.253), GB (0.207)
- **API:** `/api/emission-factors` (lookup, calculate, biogenic-correction)

16. HUM DYNAMIC LISTENER (`services/hum-listener.ts`)

- **Baseline:** 12.6 MW (replaced legacy 18.0 MW)
- **Sources:** NYSE/NASDAQ, LSE, CME/Eurex trade windows
- **Shift threshold:** 5% triggers on-chain Memo anchor
- **API:** `/api/hum-telemetry` (GET state, POST read)
- **SSE:** `/api/ticker-stream` — 3-5s interval updates for cockpit ticker

17. ADMIN CONTROL PANEL (`/admin`)

Auth Gate

Fetches `/api/auth/me`, validates FOUNDER or ADMIN role. Non-authorized users see access-denied screen.

Panels

Panel	Function
Stats Strip	Total Assets, Pending Review, Approved, Total Sovereign Value
Pending Discovery Queue	Per-asset APPROVE & SETTLE / REJECT / RE-SCAN buttons
Settled Discovery Ledger	Settled assets with settlement IDs, RE-SCAN action
Network Purge Console	Red-bordered danger zone, typed "PURGE" confirmation gate

Purge Gate (`/api/v1/admin/purge-test-artifacts`)

- GET: Non-destructive scan preview
- POST: Requires `confirmationToken: "PURGE"` — hard gate
- Dual auth: Aethexer key OR browser session JWT

18. API ROUTE INDEX

v1 Protocol Endpoints (Aethexer-authenticated)

Route	Methods	Purpose
/api/v1/forensic-ingest	GET, POST	Unified forensic file ingest + SPEXTER extension
/api/v1/telemetry-ingest	POST	IoT sensor telemetry ingest
/api/v1/telemetry-reevaluate	POST	Batch telemetry re-evaluation
/api/v1/discovery	GET, POST	Discovery asset CRUD + approval actions
/api/v1/submissions/[id]/approve	PATCH	Founder approval gate
/api/v1/admin/purge-test-artifacts	GET, POST	Destructive purge with confirmation gate

Auth Endpoints

Route	Methods	Purpose
/api/auth/login	POST	JWT login
/api/auth/register	POST	User registration
/api/auth/signup	POST	Alias for register
/api/auth/agent-signup	POST	Agent registration with invitation code
/api/auth/google	GET	Initiate Google SSO
/api/auth/callback/google	GET	Google OAuth callback
/api/auth/me	GET	Session check
/api/auth/logout	POST	Clear session

Submission Lifecycle

Route	Methods	Purpose
/api/submissions	GET, POST	List/create submissions
/api/submissions/[id]/parse	POST	AI forensic extraction
/api/submissions/[id]/settle	POST	70/20/10 settlement
/api/submissions/[id]/certificate	GET	HTML2PDF certificate generation

Domain Endpoints

Route	Purpose
/api/biochar , /api/biochar/[id] , /api/biochar/history , /api/biochar/stats	CDR engine
/api/farmers , /api/farmers/[id]	Farmer management
/api/credits , /api/credits/stats	Green credit lifecycle
/api/calculate	Asphalt carbon calculator
/api/carbon-impact , /api/carbon-score	Impact scoring
/api/vault , /api/vault/ai-check , /api/vault/destruction-proof , /api/vault/logistics-audit	Verification repository
/api/analytics , /api/stats	Dashboard analytics
/api/projects	RAP project management
/api/locations , /api/locations/[id]	Location management
/api/yards	DEPRECATED — proxies to Location model
/api/consent , /api/consent/verify , /api/consent/logs	Consent management
/api/emission-factors	Emission factor lookup
/api/spatial	PLY/Splat parsing
/api/notary	Solana notary anchor
/api/hum-telemetry	HUM listener state
/api/ticker-stream	SSE market data stream
/api/audit-logs	Audit log query
/api/invitations	Agent invitation codes
/api/agent/register-farmer , /api/agent/pending-counts , /api/agent/verification-queue	Agent portal
/api/global-ledger	Consolidated ledger data
/api/upload	S3 presigned URL generation

Route	Purpose
/api/compliance	Compliance framework

19. PAGE ROUTE INDEX (31 Pages)

Route	Purpose
/	Main dashboard + cockpit ticker
/admin	Master admin control panel (FOUNDER/ADMIN gated)
/login	Email/password + Google SSO login
/signup	User registration
/signup/agent	Agent registration with invitation code
/onboarding	3-step sovereign onboarding wizard
/submission-portal	Corporate gatehouse (wallet-gated)
/discovery	Macro-asset discovery feed
/calculator	Asphalt RAP carbon calculator
/analytics	Dashboard analytics + charts
/projects	RAP project management
/biochar	CDR calculator
/biochar/batches	Biochar batch history
/farmers	Farmer management
/credits	Green credit lifecycle
/metal	Metal recovery dashboard
/metal/calculator	Metal carbon calculator
/metal/logs	Metal recovery logs
/vault	Verification repository
/vault-node	Vault node management
/vault/history	Verification history
/vault/settings	Vault settings
/vault/community	Community view
/impact	Impact dashboard

Route	Purpose
/impact-calculator	Impact calculator
/global-ledger	Consolidated ledger
/ledger	Legacy ledger view
/legacy-healer	Friction healing calculator
/agent-portal	Agent dashboard
/agent-portal/verification-queue	One-tap verification queue

20. LIBRARY MODULE INDEX (31 Modules)

Module	Lines	Purpose
<code>universal-law.ts</code>	2,011	Zero Greed settlement engine, VAL formula, asset class routing
<code>sentinel-engine.ts</code>	303	Three-pillar scoring (95/75/75), verdict matrix, magic bytes
<code>logistics-auditor.ts</code>	231	Haversine distance, CO ₂ avoided from logistics
<code>validator.ts</code>	216	Zod schemas for ingest validation
<code>rbac.ts</code>	155	4-tier role hierarchy + 18 permissions
<code>aethexer-auth.ts</code>	146	Trinity 3-layer auth guard
<code>rbac-guard.ts</code>	~80	JWT extraction + withRbac wrapper
<code>aethexer-protocol.ts</code>	74	Protocol envelope wrapper
<code>audit-logger.ts</code>	~100	Fire-and-forget audit with SHA-256 hashes
<code>emission-factors.ts</code>	~200	ICE v4.1 + Ecoinvent (40 factors)
<code>blockchain.ts</code>	~150	Hedera + Polygon dual attestation
<code>solana-notary.ts</code>	~100	Devnet memo anchoring
<code>solana-integration.ts</code>	~80	Wallet adapter
<code>ply-parser.ts</code>	~200	PLY/Splat parser + volume estimation
<code>carbon-calculator.ts</code>	~100	Eurobitume LCA formulas
<code>metal-calculator.ts</code>	~80	Metal recovery calculations
<code>green-alpha-engine.ts</code>	~100	Green scoring
<code>rap-alpha-engine.ts</code>	~100	RAP analysis engine

Module	Lines	Purpose
legacy-healer.ts	~150	Friction healing + dead mass extraction
financed-emissions.ts	~100	PCAF financed emissions
compliance.ts	~80	Regulatory compliance checks
consent.ts	~60	SHA-256 consent hashing
vault-types.ts	~170	Vault type definitions
types.ts	~500	Core type definitions
utils.ts	~40	Formatting utilities
s3.ts	~60	AWS S3 presigned URL generation
aws-config.ts	~20	AWS client configuration
db.ts	~10	Prisma client singleton
aethex-engine.ts	~100	Aethex engine (HUM specs)
aethex-protocol.ts	~50	Legacy protocol (pre-v1)
resend.ts	~20	Email utility

21. ENVIRONMENT VARIABLES

Variable	Purpose
<code>DATABASE_URL</code>	PostgreSQL connection string
<code>JWT_SECRET</code>	JWT signing key
<code>AETHEXER_PROTOCOL_KEY</code>	Primary protocol auth key
<code>AETHEXER_FIELD_KEY</code>	Field app auth key
<code>GOOGLE_CLIENT_ID</code>	Google SSO client ID
<code>GOOGLE_CLIENT_SECRET</code>	Google SSO client secret
<code>ADMIN_MASTER_EMAIL</code>	Auto-elevation email (idecade8@gmail.com)
<code>SOLANA_NOTARY_SECRET_KEY</code>	Devnet notary keypair
<code>AWS_ACCESS_KEY_ID</code>	S3 credentials
<code>AWS_SECRET_ACCESS_KEY</code>	S3 credentials
<code>AWS_REGION</code>	S3 region
<code>AWS_BUCKET_NAME</code>	S3 bucket
<code>NEXTAUTH_URL</code>	Auto-configured per environment
<code>HTML2PDF_API_URL</code>	PDF generation service
<code>HTML2PDF_API_KEY</code>	PDF generation key

22. DEPLOYMENT

Environment	URL	Notes
Production	<code>bedrockesg.com</code>	Custom domain, LIVE
Preview	VM preview URL	Development environment

Build: Standalone output mode, rsync to `/tmp/app`, tar artifact deployed.

Database: Shared between dev and production — NO destructive operations without confirmation.

Google SSO: Scoped to `bedrockesg.com` with callback at `/api/auth/callback/google`.

